Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 36: Approved Solutions

Pascal's Triangle and the Binomial Theorem

Covers Chartrand & Zhang, Section 9.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

The three facts

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \qquad \binom{n}{r} = \binom{n}{n-r} \qquad (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r$$

For coefficient problems: write the general term first

For $(ax+by)^n$ the general term is $\binom{n}{r}(ax)^{n-r}(by)^r$. Expand the constants, match the exponent you need, solve for r, then evaluate. **Do not guess which binomial coefficient to use.**

Three mistakes to expect

1. **Forgetting the constants.** In $(3x+2y)^5$ the 3 and 2 carry powers too.
2. **Dropping the sign** in $(x-y)^n$, where the term carries $(-1)^r$.
3. **Solving for the wrong r.** Match the exponent carefully.

Part II: Practice Problems

Section A — The triangle

1.

(a) Write rows 7 and 8 of Pascal's triangle.

(b) Verify Theorem 9.1 for $\binom{8}{4}$.

(c) Verify that row 7 sums to 2^7.

Approved solution

(a) Row 7: $1, 7, 21, 35, 35, 21, 7, 1$. Row 8: $1, 8, 28, 56, 70, 56, 28, 8, 1$ — each entry the sum of the two above.
(b) $\binom{8}{4} = \binom{7}{3} + \binom{7}{4} = 35 + 35 = 70$. ✓
(c) $1 + 7 + 21 + 35 + 35 + 21 + 7 + 1 = 128 = 2^7$. ✓
This is the subset identity: row n counts the subsets of an n-set by size, and there are 2^n subsets in all.

2. Verify the Hockey Stick Theorem in the case $\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \binom{6}{3} = \binom{7}{4}$.

Approved solution

$1 + 4 + 10 + 20 = 35$, and $\binom{7}{4} = 35$. ✓

The name comes from the shape: the summed entries run down a diagonal of the triangle and the answer sits just off the end, forming a hockey stick.

Section B — Combinatorial proof

3. Prove that $\binom{n}{r} = \binom{n}{n-r}$ by a counting argument, not by algebra.

Approved solution

Let S be an n-element set. The left side counts the r-element subsets of S.
Pair each r-element subset A with its complement $S - A$, which has $n - r$ elements. This pairing is reversible: from $S - A$ you recover A by taking the complement again. So it is a bijection between the r-element subsets and the $(n - r)$-element subsets.
Since a bijection exists, the two collections have the same size, and $\binom{n}{r} = \binom{n}{n-r}$. ■

The argument is Week 7's definition of equal cardinality — exhibit a bijection — applied to finite sets. Choosing what to include and choosing what to exclude are the same act.

Section C — The Binomial Theorem

4. Find each coefficient. Write the general term first.

(a) The coefficient of x^5y^3 in $(x + y)^8$.

(b) The coefficient of x^3y^2 in $(3x + 2y)^5$.

(c) The coefficient of a^2b^5 in $(a - b)^7$.

Approved solution

(a) General term $\binom{8}{r}x^{8-r}y^r$. Need $r = 3$: $\binom{8}{3} = 56$.
(b) General term $\binom{5}{r}(3x)^{5-r}(2y)^r = \binom{5}{r}3^{5-r}2^r x^{5-r}y^r$. Need $r = 2$:

$$\binom{5}{2} \cdot 3^3 \cdot 2^2 = 10 \cdot 27 \cdot 4 = 1080.$$

(c) Write $(a-b)^7 = (a+(-b))^7$, with general term $\binom{7}{r}a^{7-r}(-b)^r = \binom{7}{r}(-1)^r a^{7-r}b^r$. Need $r=5$:

$$\binom{7}{5}(-1)^5 = 21\cdot(-1) = -21.$$

(b) and (c) are the two standard traps: constants carry powers, and subtraction carries a sign. Writing the general term first catches both.

5. Find the constant term in $\left(x+\dfrac{2}{x}\right)^6$.

Approved solution

General term:

$$\binom{6}{r}x^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{6-2r}.$$

Constant means $6-2r=0$, so $r=3$, giving $\binom{6}{3}\cdot 2^3 = 20\cdot 8 = 160$.

The exponent bookkeeping is the whole problem. The x in the denominator contributes x^{-r}, so the exponent drops by 2 each time r increases.

Section D — Writing

6. In five to eight complete sentences, explain why the coefficients in $(x+y)^n$ are the same numbers that count subsets. Describe what is being chosen when you expand the product.

Approved solution

Model answer. Expanding $(x+y)^n$ means multiplying out n identical factors of $(x+y)$, and every term in the result comes from picking either x or y from each factor and multiplying the picks together. A term equal to $x^{n-r}y^r$ arises exactly when y was chosen from r of the factors and x from the other $n-r$. So the number of such terms is the number of ways to choose which r factors contribute a y — an unordered selection of r things from n, which is $\binom{n}{r}$. Collecting like terms therefore gives $x^{n-r}y^r$ a coefficient of $\binom{n}{r}$. The same number counts r-element subsets of an n-element set because the act is identical: in both cases you are selecting r items from n without regard to order.

Full credit requires identifying what is being chosen — which factors contribute a y. An answer that only observes the coefficients match Pascal's triangle earns about half.

Section E — Challenge (optional)

7. Use the Binomial Theorem to prove that $\binom{n}{0}+\binom{n}{1}+\cdots+\binom{n}{n} = 2^n$, and then prove the same identity by a counting argument.

Approved solution

By the Binomial Theorem. Set $x = y = 1$:

$$2^n = (1+1)^n = \sum_{r=0}^{n} \binom{n}{r} 1^{n-r} 1^r = \sum_{r=0}^{n} \binom{n}{r}.$$

By counting. Count the subsets of an n-element set two ways. Grouping them by size gives $\sum_r \binom{n}{r}$. Deciding in-or-out for each element independently gives 2^n by the Multiplication Principle. Both count the same collection, so they are equal. ■

Two proofs of one identity, and they feel completely different. The first is a substitution; the second never computes anything. The counting proof also explains *why* the identity holds, which the substitution does not.
Setting $x = 1$, $y = -1$ gives another identity worth a moment's thought.

Instructor notes

- **Problem 3** is a bijection argument — Week 7's machinery used to prove a counting identity. Worth naming the connection.
- **Problems 4(b) and 4(c)** are the two standard coefficient traps. If students write the general term first, both disappear.
- **Problem 7** gives two proofs of one identity. The contrast between a substitution and a counting argument is the most valuable thing here.
- **Problem 6** is the conceptual anchor: coefficients are counts.
- Reserved for Homework 13: Section 9.1 even-numbered exercises.